ANTONIU FODOR

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EDUCATION

Ph.D. in Physics, University of Toledo Aug. 2024

Dissertation: *Major and Minor Pathways of Evolution: Suppressing Star Formation in Nearby Galaxies*

Advisor: Anne Medling

B.A. in Physics & Mathematics, Albion College May 2018

Advisors: Nicolle Zellner, David Seely

RESEARCH INTERESTS

Spectroscopy of post-starburst galaxies using optical observations to probe kinematics and ionized gas properties.

- Galactic Outflows: Identification and analysis of outflows present around shocked post-starburst galaxies to see effect on quenching signatures.
- Ionized Gas Turbulence: Probing velocity dispersion to compare internal ionized gas of post-starbursts to star-forming galaxies.
- Data Reduction Pipelines: Working with FITS files to transform raw telescope data into science-ready products.

RESEARCH EXPERIENCE

Graduate Researcher Aug. 2018 – Aug. 2025

Department of Physics and Astronomy, University of Toledo

- Led a multi-year observing campaign to collect, process, and analyze 110 shocked post-starburst galaxies using optical long-slit spectroscopy and leveraging the spatial dimension to identify outflows and radial properties.
- Investigated ionized gas properties of detected large-scale extraplanar outflows, focusing on estimating kinematics and mass information.
- Analyzed major axis ionized gas properties, radial stellar population changes, and modeled stellar ages to characterize star formation quenching in target galaxies.

Undergraduate Research Assistant Summer 2017

Physics Department, Albion College
Advisor: Nicolle Zellner
Research Focus: Photometric data reduction of eclipsing binary stars.

Undergraduate Research Assistant Summer 2016
Physics Department, Albion College
Advisor: David Seely
Research Focus: Low-energy charge exchange observations using the Ion-Atom Merged Beams Apparatus at Oak Ridge National Laboratory

ADVISING EXPERIENCE

Mary Braun 2021–2024

"Wind Signatures in Shocked Poststarburst Galaxies"

Taylor Tomko 2021–2024

"An Analysis of Active Galactic Nuclei in Shocked Poststarburst Galaxies"

Matthew Newhouse 2020–21; 2022–23

"The Impact of Stellar Absorption Features on Spectroscopic Flux Calibration"

Alexander Thompson 2019–2021

"A Spectral Comparison of Sloan Digital Sky Survey and Lowell Discovery Telescope Datasets"

Thomas Johnson 2019–2021

"Error Propagation in DeVeny Long-slit Spectroscopy"

LEADERSHIP, SERVICE & OUTREACH

Service to the Scientific Community

Spectrograph Technical Expert & Instructor Apr. 2020 - Jun. 2025

- Served as primary technical resource for Lowell Discovery Telescope operations at the University of Toledo, supporting faculty, astronomers, and students during observing sessions and authoring tutorials for telescope usage and data retrieval.
- Maintained a shared data repository for collaborative projects including documentation and tutorials, supporting 11 researchers across 5 undergraduate projects and 3 graduate research projects.
- Designed and taught data cleaning class sessions in the Astronomy 4880 course to transform raw telescope data into informative data cubes for scientific analysis, adopted as the basis for a multi-week class project.

Teaching

Graduate Teaching Assistant (University of Toledo) 2018 – 2020

Courses: General Physics I; Elementary Astronomy; Solar System Astronomy; Stars, Galaxies, and the Universe

- Instructed and supported teaching for 100+ undergraduates across STEM courses and contributed to curriculum review modernizing lab materials and improving instructional clarity.

Teaching Assistant (Albion College) 2017-2018

Outreach and Broader Impact

Volunteer, Solar Eclipse Viewing Outreach, University of Toledo 2024

Speaker, Lowell Discovery Telescope Decade of Discovery Talk Series 2023

Department Organizer, Cedar Point's Physics, Science, and Math Days Outreach 2023

Volunteer and Organizer, Solar Eclipse Viewing Outreach, Albion College 2017

PUBLICATIONS

Peer-Reviewed

Shocked POststarburst Galaxy Survey. IV. Outflows in Shocked Poststarburst Galaxies Are Not Responsible for Quenching
Fodor, A., Tomko, T., Braun, M., Medling, A., Johnson, T., Thompson, A., Johnston, V., Newhouse, M., Luo, Y., French, K.D., Otter, J.A., Tripathi, A., Verrico, M., Alatalo, K., Rowlands, K., Heckman, T., The Astrophysical Journal, Volume 979, Issue 1, article id. 94, 94 pp. (01/2025)

In Preparation

Shocked POststarburst Galaxy Survey.V. Turbulent Ionized Gas and Constant Stellar Populations
Fodor, A., Medling, A., Tomko, T., Braun, M.,A., Johnson, T., Thompson, A., Johnston, V., Newhouse, M., Luo, Y., French, K.D., Otter, J.A., Tripathi, A., Verrico, M., Alatalo, K., Rowlands, K., *in preparation for The Astrophysical Journal* (2026)

CONTRIBUTED TALKS

These Outflows are (Not) the Quenching Mechanisms You're Looking For Mar. 2026
The End of Star Formation Conference

A Minor (and Major) Search for Quenching: Optical Long-slit Spectroscopy of Shocked Post-starburst Galaxies Jan. 2025
245th American Astronomical Society Meeting

The Shocked Post-starburst Galaxy Survey: Outflows in Shocked Post-Starburst Galaxies Are Not Responsible For Quenching Jul. 2024
UIUC - Cardiff Galaxy Meeting

Looking for Triggers in the Most Recently-Quenched Galaxies May 2024
Compact Objects Meeting 2024

Looking for Triggers in the Most Recently-Quenched Galaxies Jan. 2024
243rd American Astronomical Society Meeting [iPoster]

Optical Spectroscopy of Shocked Post-Starburst Galaxies in Search of Outflow and Quenching Signatures Jun. 2022
240th American Astronomical Society Meeting

Shocked Post-starburst Galaxies: Searching for Outflows to Explain Quenching Jan. 2021
237th American Astronomical Society Meeting [iPoster]

SUCCESSFUL OBSERVING PROPOSALS

DeVeny/Lowell Discovery Telescope (Semester 2023B),
Early Warning Signs of Quenching, 5 nights

DeVeny/Lowell Discovery Telescope (Semester 2023A),
Early Warning Signs of Quenching, 6 nights

DeVeny/Lowell Discovery Telescope (Semester 2022B),
Early Warning Signs of Quenching, 6 nights

DeVeny/Lowell Discovery Telescope (Semester 2022A),
Early Warning Signs of Quenching, 6 nights

DeVeny/Lowell Discovery Telescope (Semester 2021B),
Early Warning Signs of Quenching, 6 nights

DeVeny/Lowell Discovery Telescope (Semester 2021A),
Early Warning Signs of Quenching, 6 nights

DeVeny/Lowell Discovery Telescope (Semester 2020B),
Early Warning Signs of Quenching, 6 nights

DeVeny/Discovery Channel Telescope (Semester 2020A),
Early Warning Signs of Quenching, 6 nights

OBSERVING EXPERIENCE

DeVeny/Lowell Discovery Telescope, long-slit spectroscopy - 47 nights

LMI/Lowell Discovery Telescope, optical imager - 8 nights

NIHTS/Lowell Discovery Telescope, near-infrared spectroscopy - 2 nights

RELEVANT SKILLS

Programming	Proficient in Python, SQL and IDL; Some experience in MATLAB, and Java
Data Reduction	Substantial experience in optical long-slit data reduction for ground-based instruments
Other Software	Git, Jupyter Notebook, Markdown